

Hw 01

A fast-food restaurant sells hamburgers and chicken sandwiches. On a typical weekday the demand for hamburgers is normally distributed with mean 313 and standard deviation 57; the demand for chicken sandwiches is normally distributed with mean 93 and standard deviation 22.

- a. How many hamburgers must the restaurant stock to be 98% sure of not running out on a given day?
- b. Answer part a. for chicken sandwiches
- c. If the restaurant stocks 400 hamburgers and 150 chicken sandwiches for a given day, what is the probability that it will run out of either hamburgers or chicken sandwiches or both? Assume that demand for both hamburgers and the demand for chicken sandwiches are independent.
- d. Why is the independence assumption in part c probably not realistic?
- e. Suppose that the restaurant has unlimited stocks of hamburgers and chicken. The unit margin of hamburger is 8 USD, and that of the chicken sandwich is 5 USD. Find the expectation and standard deviation of the total revenue from these two items.
 - 1) Assume both demands are independent.
 - 2) [LSCM optional]: The correlation between the two demands are $\rho = -0.37$

Hw02

Solve the following chance-constrained problem

$$\text{Maximize } z = 5x_1 + 6x_2 + 3x_3$$

subject to

$$\Pr(A_{11}x_1 + A_{12}x_2 + A_{13}x_3 \leq 8) \geq 0.95$$

$$\Pr(5x_1 + x_2 + 6x_3 \leq B_2) \geq 0.10$$

$$x_1, x_2, x_3 \geq 0$$

Assume that the parameters $A_{1j}, j = 1, 2, 3$ are independent and normally distributed random variables with the following means and variances

$$E[A_{11}] = 1, E[A_{12}] = 3, E[A_{13}] = 9$$

$$\text{var}(A_{11}) = 25, \text{var}(A_{12}) = 16, \text{var}(A_{13}) = 4$$

The parameter B_2 is normally distributed with mean 7 and variance 9